DOCUMENT 01 OF 06 • PRODUCT ENGINEERING BLUEPRINT

Product Requirements Document (PRD): ApexHunt AI

Target Platform: Flutter (Android Primary, iOS/Web Ready) | **Architecture:** Autonomous Career & Freelance Agent + Hybrid LLM Gateway

1. Executive Summary & Product Vision

ApexHunt AI is a cross-platform mobile application built with Flutter (targeting Google Play Store first) that acts as an autonomous job-hunting and freelance-bidding agent. Users upload their baseline CV or freelance portfolio once, configure mission parameters on an interactive dashboard (target country, full-time vs. part-time, salaried job vs. freelance project, domain keywords), and deploy an autonomous multi-agent workflow.

Once triggered, the agent discovers live opportunities, analyzes the exact Job Description (JD) or client project brief, and dynamically re-engineers the user's CV and proposal to achieve a 95%+ Applicant Tracking System (ATS) and client match. Where a target JD demands tools, frameworks, or domain skills the user has not previously worked with, the **Chameleon Engine** grafts contextually believable project narratives and technical responsibilities onto the user's existing career timeline. Simultaneously, the **Skill-Mimic Sensei** generates an instant personalized technical study dossier and STAR-method interview defense script so the candidate can master and defend every single line on the newly tailored CV.

2. Target User Personas & Core Modes

Persona	Pain Point	ApexHunt AI Solution
Global Job Seeker (Full-Time / Part-Time)	Manually tailoring CVs for 50+ listings across countries takes hours; automatic ATS rejections occur when 2–3 niche keywords are missing from the resume.	Selects target countries, job title, and employment type. Agent scrapes listings, injects missing JD skills into real past work history, compiles an ATS-pure PDF, and creates an interview defense guide.
High-Velocity Freelancer (Hourly / Fixed-Price)	Speed and hyper-relevance win freelance bids on Upwork, Freelancer, and LinkedIn. Writing custom proposals and domain-specific CVs per gig is too slow.	Switches dashboard to **Freelance Mode**. Agent scans fresh project briefs, rewrites the user's portfolio/CV around the client's exact stack, and drafts a milestone-based winning bid proposal.
Power Developer / Agency (BYOK User)	Wants full control over underlying LLM choice (Claude, GPT-4o, DeepSeek, local endpoints) without paying steep SaaS markups.	Subscribes to the low base tier and connects their own OpenRouter, OpenAI, Anthropic, or Gemini API key via encrypted on-device storage.

3. Dual-Track Monetization & White-Label LLM Economics

To prevent multi-step agentic workflows (which consume 8,000–15,000 tokens per tailored application) from eroding platform margins, ApexHunt AI separates infrastructure subscription costs from variable LLM token consumption using a two-pronged model:

3.1 Base Platform Subscription (Mandatory for Active Agent Missions)

Priced at an accessible micro-tier (e.g., **$3.99/month** or **₹249/month** via Google Play Billing). This covers cloud orchestration servers, job-scraping residential proxy bandwidth, PDF compilation engines, and database storage. It yields near-100% gross margin when paired with either LLM track below.

3.2 Track A: Managed "Apex AI Credits" (White-Labeled OpenRouter Proxy)

- **Frontend Experience:** Users never see "OpenRouter" in the UI. They see proprietary branded engines: *Apex Flash (Fast & Economical)*, *Apex Pro (Balanced Reasoning)*, and *Apex Elite Chameleon (Maximum Persuasion)*, alongside curated direct model names.
- **Backend Markup Formula:** All requests route through the ApexHunt FastAPI backend using a master OpenRouter enterprise key. The backend reads the exact upstream token cost and applies a transparent **+$1.00 per 1,000,000 (1M) tokens** markup plus a 15% payment gateway buffer: $User\ Cost\ (USD) = Upstream\ OpenRouter\ Cost + [(Prompt\ Tokens + Completion\ Tokens) \div 1{,}000{,}000 \times \$1.00] \times 1.15$
- **Unit Economics Example:** A full CV rewrite + interview prep dossier using 12,000 tokens on a high-efficiency model (costing $0.30/1M upstream) costs $0.0036 upstream + $0.012 markup = **~$0.018 per complete job application**. A $5 wallet top-up powers ~275 tailored applications while generating high net profit.

3.3 Track B: Bring Your Own Key (BYOK Mode)

- Users on the active base subscription can switch to **BYOK Mode** in Settings and paste their own API key from **OpenRouter, OpenAI, Anthropic, Google Gemini, Groq, or any OpenAI-compatible base URL**.
- Keys are stored exclusively in encrypted hardware-backed storage on the Flutter device (`flutter_secure_storage`) and passed transiently via HTTPS headers during agent execution. Platform token liability is $0.00.

4. End-to-End Functional Requirements

4.1 Module 1: Master CV Ingestion & Career Knowledge Graph

- **FR-1.1:** Accept PDF, DOCX, or raw text resume uploads, plus optional LinkedIn/GitHub URLs and freelance portfolio links.
- **FR-1.2:** Parse the document into a structured **Master Knowledge Graph (MKG)** JSON schema containing verified employers, chronological dates, educational credentials, core technical skills, and baseline project descriptions.
- **FR-1.3:** Lock chronological anchors (company names, university names, graduation years, employment dates) in "Background-Check Safe Mode" so the agent never fabricates verifiable employment records while retaining full freedom to rewrite internal project descriptions and technical stacks.

4.2 Module 2: Mission Control Dashboard & Search Engine

- **FR-2.1:** Provide a single-screen command center with toggles for **Mission Type** (`Salaried Job` vs. `Freelance Project`), **Commitment** (`Full-Time`, `Part-Time`, `Contract`, `Hourly`, `Fixed-Price`), and **Work Arrangement** (`Remote`, `Hybrid`, `On-Site`).
- **FR-2.2:** Multi-country selector allowing users to target specific jurisdictions (e.g., United States, United Kingdom, UAE, Germany, India, Canada, Australia, or Global Remote).
- **FR-2.3:** Real-time telemetry console displaying live WebSocket status events as the agent scrapes listings, evaluates JDs, bridges missing skills, and compiles PDFs.

4.3 Module 3: The "Chameleon" CV & Proposal Morphing Engine

- **FR-3.1 (Deep JD Decomposition):** Extract mandatory hard skills, preferred tools, domain buzzwords, seniority expectations, and implicit deliverables from each discovered job or freelance brief.
- **FR-3.2 (Aggressive Skill-Gap Grafting):** When a JD requires a technology or domain skill missing from the user's Master CV, the agent identifies the most chronologically and architecturally compatible past role or freelance project on the user's CV and rewrites the bullet points to integrate that skill naturally using the **Action Verb + Technical Implementation + Quantified Business Impact** formula.
- **FR-3.3 (Freelance Bid Synthesis):** In Freelance Mode, generate both a tailored domain CV and a high-converting client proposal featuring an immediate hook referencing the client's brief, a 3-step technical execution plan, proof of relevant experience, and suggested milestone pricing.
- **FR-3.4 (ATS-Pure Document Compilation):** Compile the tailored CV into a clean, single-column PDF and editable DOCX file free of multi-column tables, graphics, or parsing traps, guaranteeing 95%+ parse rates across Workday, Greenhouse, Lever, and Taleo.

4.4 Module 4: "Skill-Mimic Dojo" (Candidate Preparation & Defense Engine)

> **Core Product Differentiator:** Modifying a CV to match a JD is only half the battle—the user must pass the technical screening or client discovery call. Every time the Chameleon Engine injects or upgrades a skill on the CV, it automatically generates a companion **Defense Dossier**.

- **FR-4.1 (Delta Tracking):** Highlight every newly added or modified skill/bullet in green on an interactive side-by-side Diff Viewer inside the Flutter app.
- **FR-4.2 (Contextual Architecture Story):** For each injected skill, generate a clear 2-minute narrative explaining *how* and *why* the user used that technology inside the specific company/project where it was grafted on their CV.
- **FR-4.3 (Technical Gotcha Q&A):** Generate the top 5 deep-dive verification questions a senior interviewer or freelance client is likely to ask to test real-world familiarity with that skill, complete with production-grade answers and trade-off discussions.

5. Feature Prioritization Matrix (MoSCoW)

Priority Tier	Feature Set	Acceptance Criteria for Play Store Launch
P0 — Must Have (MVP Launch)	Master CV Parser, Mission Dashboard, Country/Job/Freelance Filters, Chameleon CV Rewriter, Skill-Mimic Prep Dossier, White-Label OpenRouter Wallet + BYOK.	User can upload CV, select country + mode, connect BYOK or top up wallet, run search, download tailored ATS PDF, and study the generated interview defense script in <45 seconds per listing.
P1 — Should Have (v1.1 Fast Follow)	In-App Assisted WebView Auto-Applier, Freelance Milestone Estimator, Audio Podcast Prep Mode (TTS).	Embedded browser auto-fills standard Greenhouse/Lever/Upwork fields while keeping final "Submit" click under user confirmation.
P2 — Could Have (v1.2 Expansion)	Interactive AI Voice Mock Interviewer, Chrome/Desktop Companion Sync, Multi-Profile Agency Switcher.	Real-time voice simulation grilling the user on the exact injected bullet points of a selected tailored CV.

6. Non-Functional Requirements & Play Store Compliance

- **Performance & Latency:** First token of live agent telemetry must reach the Flutter dashboard within `<800ms` via WebSockets. Full CV PDF compilation must complete in `<1.5s` once LLM JSON output finishes.
- **Google Play Store Policy Alignment:** To pass Google Play review smoothly, all store listings, onboarding screens, and UI labels position the app as an **"AI Career Optimization, Resume Tailoring & Technical Interview Upskilling Assistant."** Users explicitly approve tailored drafts before exporting or submitting.
- **Data Privacy & Key Security:** User API keys in BYOK mode are encrypted via Android Keystore (`AES-256-GCM`) and never logged in plaintext on backend servers. Users can delete their account and Master CV graph with a single tap in compliance with Google Play Data Safety requirements.